

SAN Final assignment Proposal

Endocrine disruptors and their impact on reproductive health

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1 Study overview

1.1 Research objectives

Primary objective

Examine associations between urinary concentrations of environmental phenols and phthalate metabolites with serum sex steroid hormones in the adult U.S. population. [2],[1],[3],[4].

Secondary objectives:

- Assess whether hormonal alterations mediate associations between EDC biomarkers and reproductive/fertility indicators.
- Compare combined exposures (phthalates + phenols) effects on hormones and fertility versus single chemicals.

Hypotheses:

- Higher urinary concentrations of phthalate metabolites and BPA are associated with altered serum sex steroid hormones.
- Hormonal alterations partially mediate the relationship between urinary EDCs and self-reported reproductive outcomes (e.g., infertility, menstrual irregularity).
- Mixture-based models reveal stronger or different associations than single-chemical models, and the effects differ by sex and age group.

2 Dataset and study population

2.1 Data source

National Health and Nutrition Examination Survey (NHANES) 2013-2014 cycle, available at [cdc.com](https://www.cdc.com).

2.2 Key data files

- DEMO_H: Demographics, survey design variables
- EPHPP_H: Environmental phenols and parabens
- PHTHE_H: Phthalate metabolites
- TST_H: Testosterone, estradiol
- RHQ_H: Reproductive health questionnaire (females only)
- additional: demographics, BMI, smoking, medication use, alcohol use, socioeconomic status, dietary intake

2.3 Study population

Inclusion criteria:

- Adults aged above 18
- Completed MEC examination
- At least one detectable urinary EDC measurement
- Valid subsample weight ($WTSB2YR > 0$)

Exclusion criteria:

- Pregnant at examination (for hormone analyses)
- Missing key covariates (age, sex, BMI etc.)
- Extreme urinary creatine values (>30 or > 300 mg/dL)

3 Variables

3.1 Predictors

- Urinary pththalate metabolites
- Urinary bisphenols
- Urinary parabens
- Mixture variables

3.2 Outcomes (Dependent variables)

- Serum hormone concentrations: Testosterone, estradiol, SHBG (Sex hormone-binding globulin)
- Reproductive indicators (females only)
 - Infertility (attempted pregnancy for longer than a year without success or sought medical help)
 - Ever treated for PID (pelvic inflammatory disease)
 - regular menstrual cycle (past 12 months)
 - gestational diabetes
 - hysterectomy
 - Total number of deliveries
 - age at first live birth

3.3 Covariates

- demographics (age, education, income)
- BMI
- medication use
- smoking
- alcohol use

4 Plan

1. extract and clean data
 - merge datasets by patient ID
 - remove individuals missing key exposure or outcome variables for a given analysis (criteria in 2.3)
 - variables statistical description (mean, median, std, min, max, probability distribution...), visualization, outliers
2. Feature selection
 - collinear variables
 - correlation matrix and variance inflation factors (VIF)
 - consider dimension reduction
3. Primary analyses

- Single-chemical analysis: survey-weighted multivariable linear regression of exposure/hormone, adjusted for covariates
- Fertility outcomes: survey-weighted logistic regression (binary fertility indicators)
- Use parametric (t-test) or non-parametric (Wilcoxon) to assess significant differences in means/medians between groups for each metabolite
- Use Pearson's or Spearman's correlation coefficients to compare relationships between metabolite concentrations.
- Mixture analysis: some kind of specialized regression (like WQS) to estimate combined effects of multiple chemicals on hormones and fertility indicators.
- Mediation analysis : estimate indirect effect (via hormone) and direct effect (exposure $\rightarrow$ hormone and hormone $\rightarrow$ fertility associations)

4. Secondary analysis

- sex and age stratified models
- sensitivity: creatine vs creatine-adjusted exposures, exclude participants on hormone medication, restrict to morning blood draws, correct for multiple testing
- Compare the results with other studies

References

- [1] Snigdha Alur et al. “Urinary phthalate metabolite concentrations in relation to history of infertility and use of assisted reproductive technology”. In: *Fertility and Sterility* 104.5 (Nov. 2015), pp. 1227–1235. ISSN: 0015-0282. DOI: 10.1016/j.fertnstert.2015.07.1150.
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- [4] Wenqiang Zhan et al. “Association between co-exposure to phenols and phthalates mixture and infertility risk in women”. In: *Environmental Research* 215 (2022), p. 114244. ISSN: 0013-9351. DOI: <https://doi.org/10.1016/j.envres.2022.114244>. URL: <https://www.sciencedirect.com/science/article/pii/S0013935122015717>.